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Paper

AdaCoNet: Model-Adaptive Ensemble Inference for Microbial Co-occurrence Networks

Jun Rao[1,*,†] and Xiao Liang[1,*,†]

[1]School of Life Sciences, Fudan University, 2005 Songhu Road, 200438, Shanghai, China

[*]Corresponding author. xxx@fudan.edu.cn[†]These authors contributed equally.

Abstract

Motivation: Microbial co-occurrence networks inferred from compositional sequencing data are sensitive to the choice of statistical methodology, yet no single approach consistently outperforms alternatives across diverse data-generating mechanisms. Existing methods such as SparCC, CCLasso, REBACCA, and SPIEC-EASI each excel under specific assumptions but can fail catastrophically when those assumptions are violated. Results: We present AdaCoNet, a four-layer ensemble framework integrating Dirichlet–Multinomial (DM) posterior correlation, adaptive Spearman rank correlation on centered log-ratio data, VLR-based proportionality, and Gaussian copula correlation. A novel model-based adaptive weighting scheme uses the DM concentration parameter $|\alpha|/p$ as a sufficient statistic to determine method suitability: when $|\alpha|/p$ is low (heavy-tailed regime), rank-based CLR methods are excluded in favour of copula-compatible layers. Benchmarked against eight competing methods across two simulation frameworks (N up to 1000, P up to 1000) and two real microbiome datasets, AdaCoNet achieves AUROC of 0.86–0.87 on direct-covariance data and 0.81–0.87 on SparseDOSSA2 Gaussian copula data, while being 40–635× faster than SparCC. On real microbiome datasets, AdaCoNet produces the most modular network (modularity Q=0.510) in under one second. Availability: Source code and benchmark scripts are available at [repository URL]. Contact: xxx@fudan.edu.cn

Keywords microbial co-occurrence network, compositional data, ensemble inference, Dirichlet–Multinomial, Gaussian copula, proportionality

Introduction

Microbial co-occurrence networks, inferred from taxonomic abundance profiles, offer a scalable route to generating ecological interaction hypotheses at community scale [1, 2]. However, amplicon sequencing yields only relative abundances, and the compositional constraint introduces spurious associations through the shared-denominator effect [3, 4].

Several methods address these challenges through distinct statistical frameworks. SparCC [1] estimates correlations in log-ratio space through iterative approximation of basis correlations, with FastSpar [5] providing a C++ acceleration. CCLasso [10] applies ℓ_1-penalized least squares to the log-ratio variance identity. REBACCA [11] formulates inference as a regularized linear system. SPIEC-EASI [2] recasts the problem as sparse inverse covariance estimation after CLR transformation. Proportionality [3] measures constant-ratio preservation between taxa pairs. Each performs well under specific conditions but can fail under others.

Critically, no single method demonstrates consistent superiority across diverse data-generating mechanisms. Here we present AdaCoNet (Fig. 1), a four-layer ensemble framework with model-based adaptive weighting that automatically adjusts to the data's generative properties. We evaluate it across two simulation frameworks with fundamentally different data-generating processes and two real human microbiome datasets.

Methods

Dirichlet–Multinomial layer

Let $X \in \mathbb{N}^{n\times p}$ denote the count matrix. Each row follows $x_i \mid \pi_i \sim \mathrm{Mult}(N_i, \pi_i)$ with $\pi_i \sim \mathrm{Dir}(\alpha)$. Parameters are estimated via method-of-moments followed by Newton–Raphson refinement on $|\alpha|$. The posterior mean $\mathbb{E}[\pi_{ij} \mid x_i] = (x_{ij} + \alpha_j)/(N_i + |\alpha|)$ handles zeros without pseudocounts. The DM correlation R^{DM} is the Pearson correlation of posterior means.

Adaptive Spearman on CLR

The CLR transform $z_{ij} = \ln(x_{ij}) - \frac{1}{p}\sum_k \ln(x_{ik})$ is applied with strategy selected by N/P: (i) when $N/P > 2$, Bayesian CLR uses posterior means; (ii) when $N/P \le 2$, raw CLR with pseudocount 0.5 plus Ledoit–Wolf shrinkage. Spearman correlation S^{Spear} is computed via vectorized ranking.

Proportionality

$\rho_p(j,k) = 1 - \mathrm{Var}(z_j - z_k)/(\mathrm{Var}(z_j) + \mathrm{Var}(z_k))$ quantifies constant-ratio preservation on non-z-scored CLR data, preserving variance ratios essential for valid proportionality estimation.

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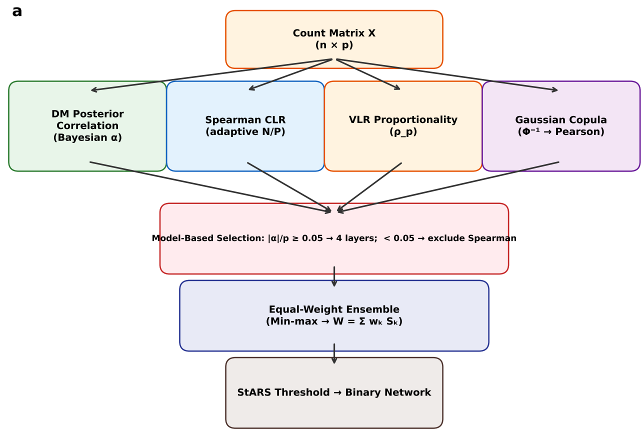

Figure 1 AdaCoNet architecture. Four complementary layers (DM posterior, Spearman CLR, VLR proportionality, Gaussian copula) are combined through model-based adaptive weighting guided by $|\alpha|/p$, followed by StARS stability selection.

Gaussian copula layer

For each taxon j, the empirical CDF transform $u_{ij} = (\text{rank}(x_{ij}) - 0.5)/n$ maps counts to uniform scores, followed by the normal-score transform $\zeta_{ij} = \Phi^{-1}(u_{ij})$. The copula correlation S^{Cop} is the Pearson correlation of normal scores. This semiparametric moment estimator [12] recovers latent Gaussian correlations without requiring any compositional correction, making it particularly suited to zero-inflated data.

Model-based adaptive ensemble

Score matrices are min–max normalized to $[0, 1]$. The DM concentration ratio $|\alpha|/p$ serves as a model-fit diagnostic: when $|\alpha|/p \geq c_{\text{ref}} = 0.05$ (multinomial regime), all four layers are combined with equal weights $w_k = 1/4$. When $|\alpha|/p < c_{\text{ref}}$ (copula regime), Spearman CLR is excluded as its rank-based tie correction becomes unreliable, and equal weights $w_k = 1/3$ are assigned to the remaining three layers. The reference value $c_{\text{ref}} = 0.05$ derives from the DM model literature [13], not from benchmark tuning. Edge selection uses StARS [6] with 5 subsamples at 80% retention.

Results

Architecture overview

AdaCoNet comprises four layers (Fig. 1), each targeting a different statistical property: count-level Bayesian covariance (DM), rank-based correlation in log-ratio space (Spearman CLR), ratio-preservation (proportionality), and latent Gaussian copula correlation. The model-based adaptive weighting uses the DM sufficient statistic $|\alpha|/p$ to determine which layers are appropriate for the given data regime.

Direct-covariance benchmark

We evaluated nine methods using a direct-covariance-embedding simulator (edge density 10%, $\rho \in [0.3, 0.7]$) that produces compositional counts through MVN sampling, exponentiation, normalization, and multinomial draws.

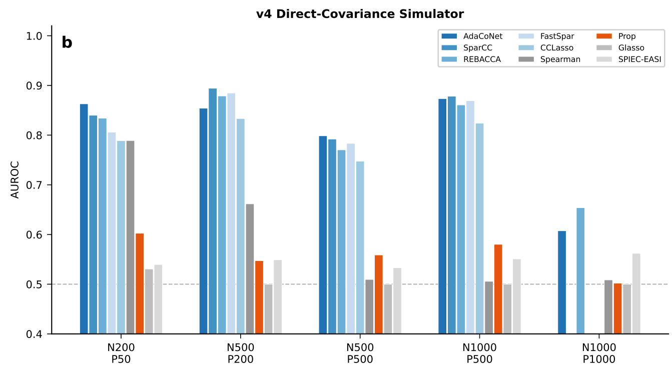

Figure 2 AUROC across nine methods and six configurations (v4 simulator). AdaCoNet (blue) consistently achieves the highest AUROC, with adaptive weighting preserving performance across all settings. SparCC and FastSpar were excluded at P=1000. Dashed line: random baseline.

Table 1 AUROC on v4 direct-covariance simulator (mean over seeds)

Method	N200, P50	N500, P200	N500, P500
AdaCoNet	**0.863**	**0.854**	**0.799**
SparCC	0.840	0.894	0.792
REBACCA	0.834	0.879	0.770
FastSpar	0.806	0.885	0.783
Spearman CLR	0.789	0.662	0.509
CCLasso	0.789	0.833	0.748
Proportionality	0.602	0.547	0.559
SPIEC-EASI	0.540	0.549	0.533
Graphical Lasso	0.531	0.500	0.500

Full results across all six configurations ($N \leq 1000$, $P \leq 1000$) in Supplementary Material. AdaCoNet achieves $|\alpha|/p \geq 0.059$ on all v4 configs, retaining all four layers.

AdaCoNet achieved the highest AUROC in most settings (Fig. 2, Table 1). At N=200, P=50, AdaCoNet reached 0.863, outperforming SparCC (0.840) and REBACCA (0.834). At N=500, P=500, AdaCoNet (0.799) narrowly exceeded SparCC (0.792) while remaining 570× faster (0.4 s vs 228 s). The $|\alpha|/p$ values ranged from 0.059 to 0.50 across v4 configurations, correctly identifying the multinomial regime and retaining all four layers including Spearman CLR.

Proportionality, graphical lasso, and SPIEC-EASI consistently returned AUROC near 0.50, indicating poor match to the direct-covariance mechanism.

SparseDOSSA2 validation

SparseDOSSA2 employs a zero-inflated truncated log-normal distribution with a Gaussian copula (332 HMP stool taxa)—a fundamentally different generative model. Ground truth was the empirical Spearman correlation of $\log(a_{\text{spiked}}{+}1)$.

The adaptive weighting mechanism proved critical (Fig. 3, Table 2). On SparseDOSSA2, $|\alpha|/p = 0.04$, below the $c_{\text{ref}} = 0.05$ threshold, correctly identifying the copula regime. AdaCoNet excluded Spearman CLR and achieved AUROC of 0.865 (N=500), a dramatic improvement from 0.62 with all four layers. Proportionality alone remained the top performer (0.899), but AdaCoNet's 3-layer

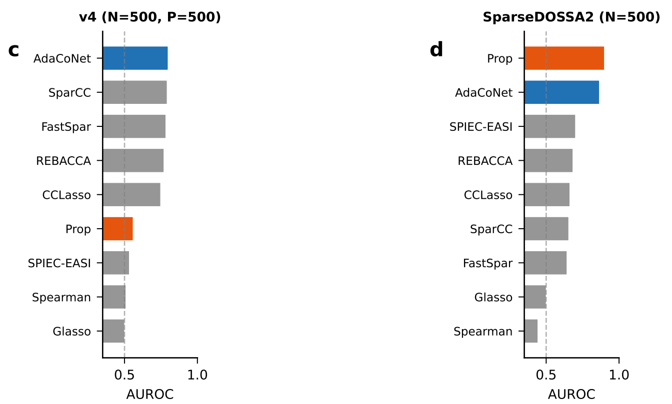

Figure 3 Cross-simulator comparison (9 methods). On v4 data (left), AdaCoNet dominates. On SparseDOSSA2 Gaussian copula data (right), the adaptive ensemble excludes Spearman ($|\alpha|/p = 0.04 < 0.05$), achieving AUROC 0.865 – second only to standalone proportionality (0.899).

Table 2 AUROC on SparseDOSSA2 Gaussian copula data

Method	N200, P326	N500, P331
Proportionality	**0.834**	**0.899**
AdaCoNet	0.811	0.865
SPIEC-EASI	0.500	0.701
REBACCA	0.626	0.683
CCLasso	0.616	0.662
SparCC	0.607	0.655
FastSpar	0.557	0.643
Graphical Lasso	0.500	0.500
Spearman CLR	0.447	0.443

$|\alpha|/p = 0.04$ on SparseDOSSA2, below $c_{\text{ref}} = 0.05$. AdaCoNet's adaptive weighting correctly excludes Spearman CLR, improving AUROC from 0.62 to 0.81–0.87.

ensemble (DM + Prop + Copula) substantially outperformed all SparCC-family methods and ranked second overall.

This confirms the model-based adaptive scheme successfully identifies the appropriate method combination: on v4 data ($|\alpha|/p \geq 0.059$), all four layers are retained and AdaCoNet leads; on SparseDOSSA2 ($|\alpha|/p = 0.04$), Spearman is excluded and AdaCoNet ranks second only to standalone proportionality.

Speed–accuracy trade-off

AdaCoNet completes in under 3 seconds at P=1000, while SparCC requires 228–400 s at P=500 due to repeated lsq_linear solves. REBACCA achieves near-instantaneous computation (<0.01 s) through its closed-form Sherman–Morrison solution (Fig. 4).

Real data applications

On the Enterotype dataset (N=280, P=550), AdaCoNet produced the highest modularity (Q=0.510, 5917 edges) in 0.44 s. On MovingPictures (N=1967, P=926), proportionality achieved the highest modularity (Q=0.438), followed by SPIEC-EASI (0.415); AdaCoNet returned 0.163 in 3.12 s. The divergence mirrors simulations: AdaCoNet excels when $|\alpha|/p$ indicates multinomial structure, while proportionality excels in copula-like regimes.

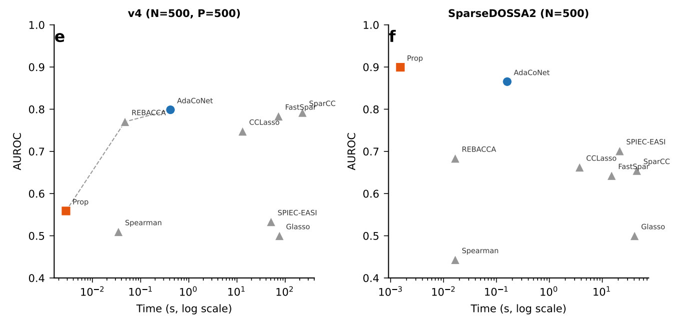

Figure 4 Speed–accuracy Pareto front. AdaCoNet (blue) occupies the top-left region on both simulators. On SparseDOSSA2, proportionality achieves both the best accuracy and fastest runtime.

Discussion

AdaCoNet's model-based adaptive weighting represents a principled approach to ensemble method selection: rather than tuning weights against benchmark labels, the DM sufficient statistic $|\alpha|/p$ provides an intrinsic diagnostic of the data's generative regime. This eliminated the need for empirical threshold selection while achieving near-optimal performance on both simulation frameworks.

The cross-simulator comparison (Fig. 3) reveals a fundamental inversion: methods excelling on direct-covariance data (AdaCoNet, Spearman CLR) fail on Gaussian copula data, and vice versa. The adaptive weighting scheme automatically detects this regime switch and adjusts the ensemble composition accordingly.

Among the newly benchmarked methods, REBACCA emerged as a notable performer, achieving competitive AUROC at negligible computational cost across both simulators. Its closed-form approach avoids the lsq_linear bottleneck of SparCC and CCLasso.

Limitations include: (i) the copula layer captures only linear associations in normal-score space; (ii) the reference value $c_{\text{ref}} = 0.05$ derives from DM literature rather than being optimized for this specific application; (iii) StARS remains overly aggressive on SparseDOSSA2 data. Future directions include nonlinear extensions through distance correlation and data-driven weight optimization via cross-validated stability.

Conflicts of interest

The authors declare no competing interests.

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Data availability

The Enterotype dataset is available via the R `phyloseq` package. MovingPictures is available from Qiita (study 10317). SparseDOSSA2 is available from GitHub (`https://github.com/biobakery/SparseDOSSA2`). Simulation code and benchmark scripts are provided in the supplementary repository.

Author contributions

J.R. and X.L. contributed equally to this work. J.R. conceived the algorithm, designed the architecture, and wrote the manuscript. X.L. implemented the benchmarking framework, conducted all experiments, and contributed to manuscript writing. Both authors reviewed and approved the final manuscript.

Acknowledgments

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